EXPERIMENT:-5

PROBLEM STATEMENT:-

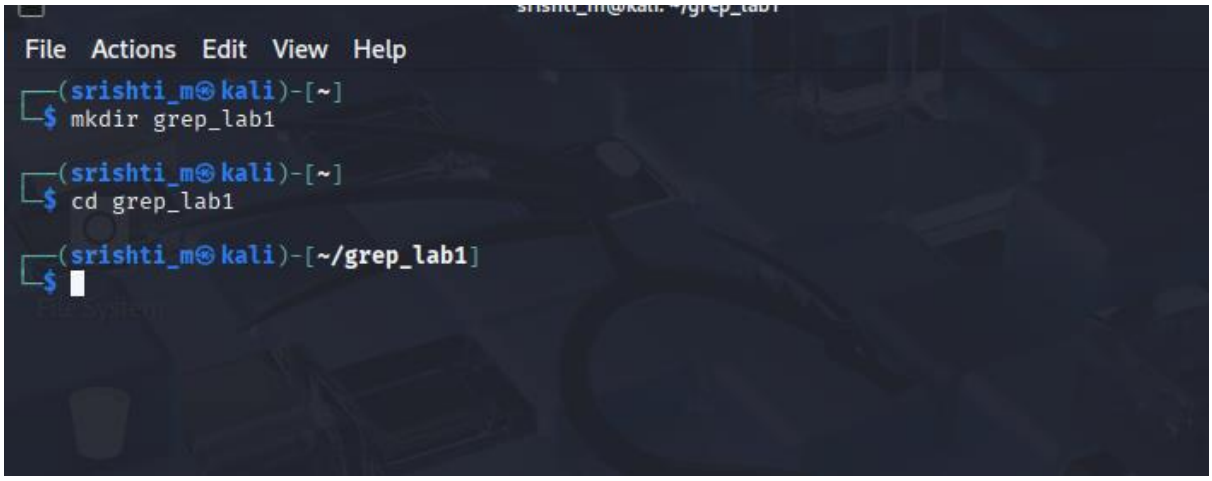
In Linux systems, large volumes of data are often stored in text files such as logs, configuration files, and scripts. Manually searching for specific information inside these files is inefficient and error-prone. Linux provides powerful text-searching utilities that enable users to quickly locate patterns, keywords, or strings within files and command outputs.

The problem is to search for specific patterns and text within files using Linux commands such as grep, fgrep, and related options, and to understand their importance in log analysis, troubleshooting, and automation tasks.

Aim :- Deleting, Copying, Moving and Renaming files in linux. Searching for file and text using grep and related.

Procedure:-

1. Log in to the Linux system and open the terminal.
   • Create a working directory: mkdir grep_lab , cd grep_lab



2.Create a sample text file: nano sample.txt

```
srishti_m@kali: ~/grep_lab1
File  Actions  Edit  View  Help
  GNU nano 8.4                    sample.txt *
linux is powerful.
linux is secure.
cybersecurity use linux.
```

3.Display file contents: cat file1.txt

```
┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$ cat sample.txty
linux is powerful.
linux is secure.
cybersecurity use linux.

┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$
```

4.Search for a word using grep: grep Linux sample.txt

```
┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$ grep linux sample.txty
linux is powerful.
linux is secure.
cybersecurity use linux.

┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$
```

5.Perform case-insensitive search: grep -i linux sample.txt

```
┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$ grep -i linux sample.txty
linux is powerful.
linux is secure.
cybersecurity use linux.

┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$
```

6. Display line numbers with matches: grep -n Linux sample.txt

```
┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$ grep -n linux sample.txty
1:linux is powerful.
2:linux is secure.
3:cybersecurity use linux.

┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$
```

7. Invert the match: grep -v Linux sample.txt

```
┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$ grep -v linux sample.txty


┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$
```

8. Search using fgrep: fgrep "cybersecurity" sample.txt

```
┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$ fgrep "cybersecurity" sample.txty
cybersecurity use linux.

┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$
```

9. Perform recursive search (if multiple files exist): grep -r Linux .

```
┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$ grep -r linux
sample.txty:linux is powerful.
sample.txty:linux is secure.
sample.txty:cybersecurity use linux.

┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$
```

Expected Output

• Matching lines containing the search pattern are displayed.

• Line numbers are shown where applicable.

• Case-insensitive and inverted searches work correctly.

Result

Text searching operations were successfully performed using grep and related commands. The student gained practical experience in locating specific patterns within files, which is essential for log analysis and automation tasks.